



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd. Razia Alamgir Banu
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Roll Number	2520090071
Branch	CSIT
Course Outcome	COURSE OUTCOME - 3

Case Study Topic: HDFC NetBanking — daily-spend prefix sums for monthly statements.

Work Done Summary

This case study analyzes how HDFC NetBanking manages daily customer spending records for monthly statements. Since transactions are updated frequently and customers often request total spending over a date range, a Binary Indexed Tree (BIT) is used. The BIT efficiently supports point updates and range-sum queries in $O(\log n)$ time while using minimal memory.

Implementation Details

1. Implemented a Binary Indexed Tree (Fenwick Tree) to store daily spending amounts.
2. Supported point updates whenever a new transaction occurs.
3. Used prefix sums to calculate spending over a range of days.
4. Processed range-sum queries using two prefix sum calculations.
5. Achieved efficient $O(\log n)$ updates and queries.
6. Required only $O(n)$ memory for storage.
7. Suitable for high-frequency banking transactions and statement generation..

Output

```
Days Recorded : 15

Query Spend [5,12]
Total Spend : 9100

Point Updates : Successful
Range Sum Query : Successful

Process finished with exit code 0
```

Result

1. Successfully maintained daily spending records.
2. Computed monthly spending ranges efficiently.
3. Supported frequent transaction updates in real time.
4. Reduced query processing time using prefix sums.
5. Demonstrated efficient handling of banking transaction data.

Time Complexity

Operation	Binary Indexed Tree
Build Tree	$O(n \log n)$
Point Update	$O(\log n)$
Prefix Sum Query	$O(\log n)$
Range Sum Query	$O(\log n)$

Conclusion

The Binary Indexed Tree proved to be an efficient solution for managing daily spending records in online banking systems. It supports fast point updates and range-sum queries with $O(\log n)$ complexity while using minimal memory. Therefore, it is well suited for generating monthly statements and processing high-frequency banking transactions.

GitHub Link: <https://github.com/vigneshkumar-stack/DSA-Project>

Student Signature	Faculty Signature